

Networking Testing Practical

1. Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other.

First, I connected my laptop and my friend's phone to the **same Wi-Fi network**. I checked the phone's IP address from its Wi-Fi settings. Then, I opened **Command Prompt** on my laptop and used the ping command with the phone's IP address.

Command:

```
ping 192.168.1.5
```

If I received **Reply from 192.168.1.5**, it means both devices can communicate with each other over the Wi-Fi network. If I got **Request timed out**, communication may be blocked by the phone's firewall or Wi-Fi network settings.

Result: The ping test successfully verified communication between the laptop and the phone.

2. Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.

I used the tracert command on Windows to trace the route to www.youtube.com.

Command:

```
tracert www.youtube.com
```

The trace reached the destination in **5 hops**.

Hop	Result
1	Request timed out
2	2402:a00:404:1::1111
3	2402:a00:1::3d
4	2402:a00::a
5	2001:4860:4826:400:: — Destination

Conclusion: The traceroute successfully reached YouTube, and the destination was reached after **5 hops**. The timeout at hop 1 does not necessarily indicate a problem; that device simply did not respond to the traceroute request.

3. Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.

I disconnected my friend's phone from the Wi-Fi network and then tried to ping the phone again from my laptop.

Command:

```
ping 192.168.1.5
```

Since the phone was disconnected, the laptop could not reach it. The error message I received was:

Request timed out.

Conclusion: The ping failed because the phone was no longer connected to the same Wi-Fi network. This shows that both devices need to be connected to the network for them to communicate with each other.

4.Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.

I used the arp -a command on my Windows laptop to view the IP and MAC addresses of devices on my local Wi-Fi network.

IP Address	MAC Address	Device / Gadget
192.168.1.1	A8-02-DB-84-94-7A	Wi-Fi Router
192.168.1.2	34-E8-94-2A-8F-D9	Connected Device
192.168.1.255	FF-FF-FF-FF-FF-FF	Broadcast Address

The addresses 224.0.0.22, 224.0.0.251, 224.0.0.252, 239.255.255.250, and 255.255.255.255 are **multicast/broadcast addresses**, not individual devices.

Conclusion: The arp -a command showed two dynamically learned devices on my Wi-Fi network: the router (192.168.1.1) and another connected device (192.168.1.2). The exact identity of 192.168.1.2 cannot be determined from the ARP output alone; it can be identified by checking the device's IP/MAC address in its Wi-Fi settings.

5.Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed.

Network Setup and Connectivity Testing Report

Introduction

The purpose of this practical was to set up a simple network using a laptop and a friend's phone connected to the same Wi-Fi network. I then tested network connectivity using the ping, tracert, and arp -a commands in Windows Command Prompt.

1. Network Setup

First, I connected my laptop and my friend's phone to the **same Wi-Fi network**. After connecting both devices, I checked the IP address of the devices. My laptop was connected to the network with the IP address **192.168.1.6**.

To check whether the two devices could communicate, I used the ping command from my laptop and entered the IP address of my friend's phone.

ping <phone-IP-address>

When the phone was connected to the same Wi-Fi network, the ping test could be used to check whether the laptop could reach the phone.

2. Testing Connectivity with Traceroute

Next, I tested the route from my laptop to YouTube using the Windows `tracert` command.

`tracert www.youtube.com`

The command successfully reached the destination in **5 hops**. The results were:

```
1  *      *      *      Request timed out.
2  1 ms    1 ms    1 ms    2402:a00:404:1::1111
3  1 ms    1 ms    1 ms    2402:a00:1::3d
4  12 ms   11 ms   12 ms   2402:a00::a
5  12 ms   12 ms   12 ms   2001:4860:4826:400::
```

The first hop showed **Request timed out**, but the following hops responded normally and the destination was reached successfully. Therefore, the timeout at the first hop did not prevent communication with YouTube.

3. Testing ARP Information

I also used the `arp -a` command to view the IP and MAC addresses known by my laptop.

`arp -a`

The important entries from my result were:

IP Address	MAC Address	Identification
192.168.1.1	A8-02-DB-84-94-7A	Wi-Fi Router
192.168.1.2	34-E8-94-2A-8F-D9	Connected Device

Other addresses shown by the command, such as 224.0.0.22 and 255.255.255.255, were multicast or broadcast addresses rather than individual devices.

4. Testing After Disconnecting a Device

Finally, I disconnected one device from the Wi-Fi network and tried to ping it again from my laptop. The ping failed and displayed:

Request timed out.

This showed that the laptop could no longer communicate with the disconnected device.

Conclusion

The practical helped me understand basic network connectivity and troubleshooting. I learned how to use **ping** to test communication between devices, **tracert** to find the route to a remote destination, and **arp -a** to view IP and MAC address information on the local network. The traceroute successfully reached YouTube in **5 hops**, while the disconnected device produced a **Request timed out** message when I tried to ping it.